

Sunghyun Lee

Student

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Summary

I am a Master's student at the Kim Jaechul Graduate School of AI, KAIST, advised by Professor Yong Man Ro. My research focuses on Multimodal AI and Computational Creativity.

Education

Yonsei University, Seoul, Republic of Korea

Computer Science & French Language and Literature (Double Major, Summa Cum Laude)

Bachelor

March 2020 - August 2025 | GPA: 3.95/4.3

Korean Advanced Institute of Science and Technology (KAIST), Daejeon, Republic of Korea

Artificial Intelligence

Master

March 2026 - February 2028

Work Experience

KAI SYSTEM | Machine Learning Engineer & Research Assistant Intern

February 2024 - November 2025

Developed a Machine Learning pipeline for KFEMS (Korean Factory Energy Management System) handling sequential industrial data.

- Built and experimented with AI pipelines using industrial data
- Patent registered in the Republic of Korea for a real-time energy data management method
- Copyright registration pending

MindForge | Research Assistant Intern

December 2023 - February 2024

Developed a Real-time 3D hand pose estimation model for HCI.

Republic of Korea Army | Mandatory Military Service

February 2021 - August 2022

Completed mandatory military service in the Republic of Korea.

Publications

Do Language Models Associate Sound with Meaning? A Multimodal Study of Sound Symbolism |

Association for the Advancement of Artificial Intelligence (November 2025)

Jinhong Jeong*, **Sunghyun Lee***, Jaeyoung Lee, Seonah Han, and Youngjae Yu

Accepted in AAAI 2026 as Oral Presentation as co-lead author

MAVL: A Multilingual Audio-Video Lyrics Dataset for Animated Song Translation |

Empirical Methods in Natural Language Processing (May 2025)

Woohyun Cho, Youngmin Kim, **Sunghyun Lee**, and Youngjae Yu

Accepted in EMNLP 2025 Main Conference as co-second author

Speaking Beyond Language: A Large-Scale Multimodal Dataset for Learning Nonverbal Cues from

Video-Grounded Dialogues | Association for Computational Linguistics (June 2025)

Youngmin Kim*†, Jiwan Chung*, Jisoo Kim, **Sunghyun Lee**, Sangkyu Lee, Junhyeok Kim, Cheoljong Yang, and Youngjae Yu

Accepted in ACL 2025 Main Conference as third author

Projects

Implementing RAG LLM Service for Korean Factory Energy Management System (KFEMS) (August 2025 - November 2025)

Developed a RAG LLM service for Korean Factory Energy Management System (KFEMS) to provide energy management insights and recommendations.

- Implemented the prototype of internal RAG LLM service for FEMS.

Awards

Graduated with Summa Cum Laude | Yonsei University (August 2025)

Graduated with Top 3% of the college in GPA

Yonsei GenAI contest 2024, silver prize | Yonsei University (December 2024)

AI-Powered Desktop PC Recommendation System with Conversational Interface

Yonsei University Dean's List | Yonsei University (February 2021)

Dean's List for the academic year 2020 fall semester

Yonsei University Dean's List | Yonsei University (February 2023)

Dean's List for the academic year 2022 fall semester

Skills

Computer Science: Python, PyTorch, TensorFlow, Scikit-learn, Computer Vision, Natural Language Processing, Machine Learning, Deep Learning, Data Science, Data Preprocessing and Construction

French: French linguistics (Pragmatics, Semantics, Syntax, Morphology), French literature (Video art, Poetry, Novel)

Languages

Korean (Native speaker), English (Fluent (Full Professional Proficiency)), French (Intermediate), Japanese (Basic)

Certificates

Microsoft Certified: Azure AI Fundamentals | Microsoft (August 2023)

Military Service Completion | Republic of Korea Army (August 2022)